The CDC **Behavioral Risk Factor Surveillance System (BRFSS)** dataset contains **284,142 observations and 33 variables**, covering data from 2015 to 2022. The dataset contains information related to older-adult health, including cognitive decline, mental health, physical health, nutrition, physical activity, smoking, caregiving, and preventive health measures.

General Information

Characteristic	**Value**
Observations	284,142
Variables	33
Years	2015–2022
Locations	59
Health classes	7
Topics	39
Data source	BRFSS
Cognitive Decline observations	22,182
SCD-specific observations	5,854

Data Type and Variables

Data type	**Number of variables**
object	24
int64	3
float64	6

Total	**33**

All Values

	Data Type	Unique Values	Missing Count	Missing Percentage
row id	object	36046	0	0.00
year start	int64	8	0	0.00
year end	int64	8	0	0.00
locationa bbr	object	59	0	0.00
location desc	object	59	0	0.00
data source	object	1	0	0.00
class	object	7	0	0.00
topic	object	39	0	0.00
question	object	39	0	0.00
data_value_unit	object	2	0	0.00
data value type id	object	2	0	0.00
data_value_type	object	2	0	0.00
data_value	float64	1001	91334	32.14
data_value_alt	float64	1001	91334	32.14
low_confidence_limit	float64	995	91545	32.22
high_confidence_limit	float64	988	91545	32.22
stratification category 1	object	1	0	0.00

stratification 1	object	3	0	0.00
stratification category 2	object	2	36873	12.98
stratification 2	object	7	36873	12.98
geolocation	object	54	30489	10.73
class id	object	7	0	0.00
topic id	object	39	0	0.00
question id	object	39	0	0.00
location id	int64	59	0	0.00
stratification category id 1	object	1	0	0.00
stratification id 1	object	3	0	0.00
stratification category id 2	object	3	0	0.00
stratification id 2	object	8	0	0.00
:@computed_region_skr5_azej	float64	52	35642	12.54
:@computed_region_hjsp_umg2	float64	51	39356	13.85
data_value_footnote_symbol	object	5	174166	61.30
data_value_footnote	object	5	174166	61.30

Total cells: 9,376,686
Missing cells: 893,323
Overall missing percentage: 9.53 %

Missing Values

Variable	Missing	Percentage

data_value_footnote_symbol	174,166	**61.30%**
data_value_footnote	174,166	**61.30%**
high_confidence_limit	91,545	**32.22%**
low_confidence_limit	91,545	**32.22%**
data_value	91,334	**32.14%**
data_value_alt	91,334	**32.14%**
stratification2	36,873	**12.98%**
stratificationcategory2	36,873	**12.98%**
:@computed_region_hjsp_umg2	39,356	**13.85%**
:@computed_region_skr5_azej	35,642	**12.54%**
geolocation	30,489	**10.73%**

Although the dataset has an overall missingness rate of 9.53%, the primary outcome variable, **data_value**, has a substantially higher missingness rate of 32.14%. This missingness must be considered during data cleaning and statistical analysis because **data_value** represents the reported health estimate used for subsequent analyses.

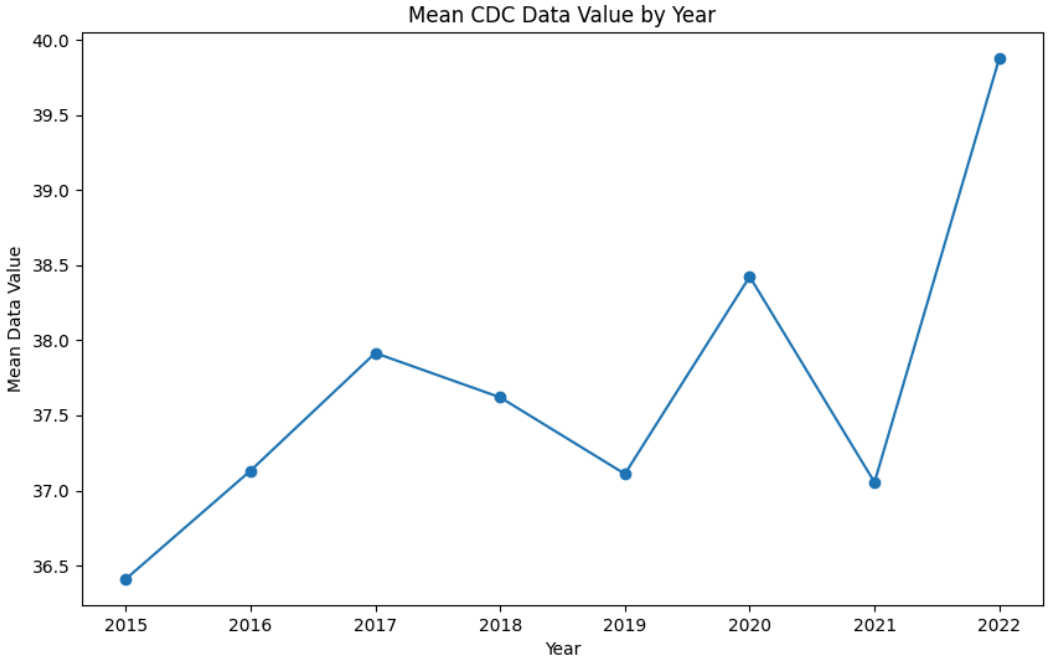


Figure 1. Mean CDC Data Value by Year (2015–2022). Mean reported data values varied across the study period, ranging from approximately 36.4 in 2015 to 39.9 in 2022. The highest mean was observed in 2022, while the lowest occurred in 2015.

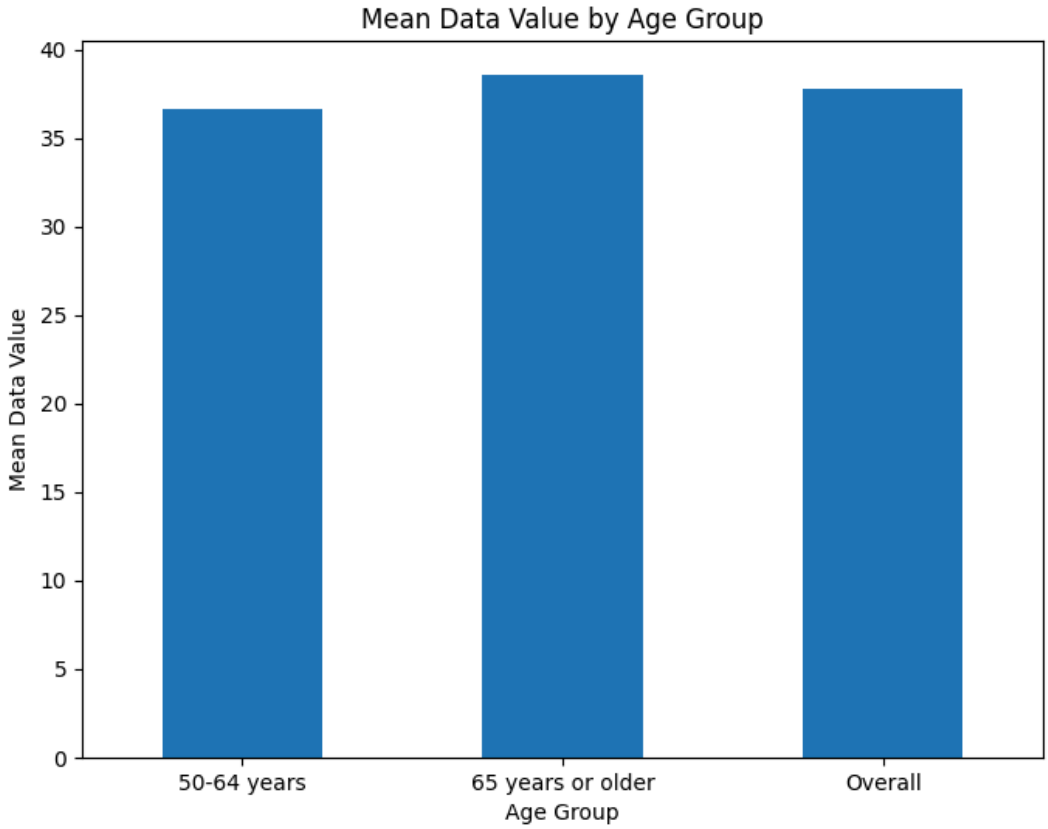


Figure 2. Mean CDC Data Value by Age Group. The mean data value was highest among adults aged 65 years or older (38.58), followed by the overall population (37.80) and adults aged 50–64 years (36.69).

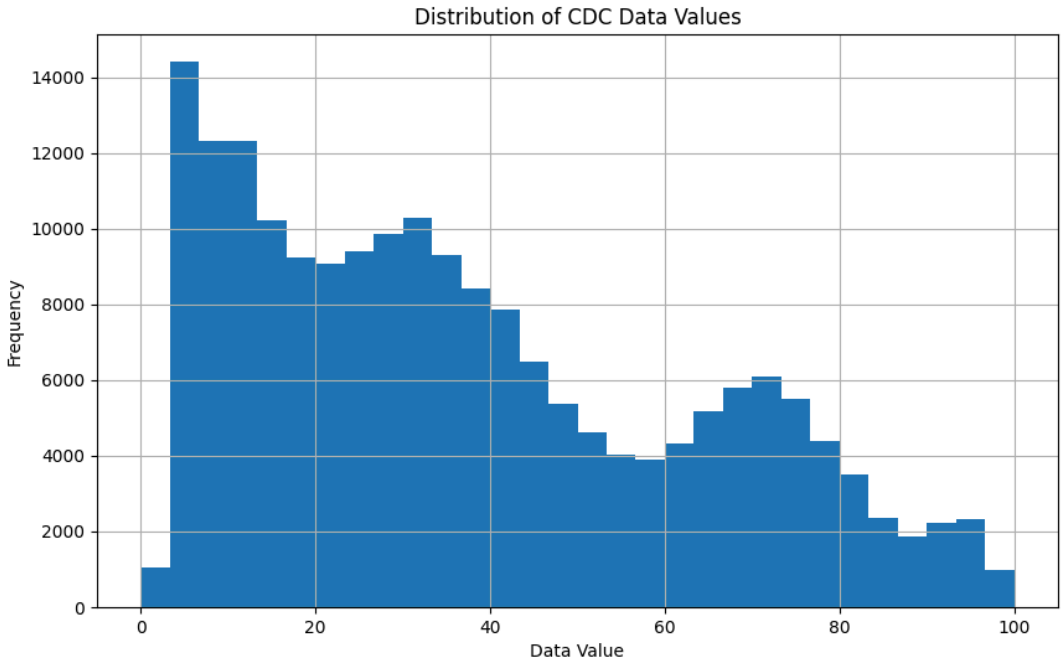


Figure 3. Distribution of CDC Data Values. The distribution of reported data values ranges from 0 to 100 and is concentrated across several value ranges, with a larger number of observations occurring at lower values. The distribution is not approximately normal and shows multiple areas of concentration.

overall **data_value** statistics are:

Statistic	**Value**
Mean	37.68
Median	32.8
Standard deviation	25.21
Minimum	0
Maximum	100

Measure Type	Mean	Median	Standard Deviation	Minimum	Maximum
Percentage measures	40.40	35.40	24.41	0	100
Mean-based measures	5.84	5.70	1.44	1.0	20.5

Across the percentage-based health measures, the average reported value was **40.40%**, with substantial variation across measures. The mean-based measures had an average value of **5.84**, representing quantities such as average days with health limitations.

Descriptive statistics by health topic

Health Topic	Count	Mean	Median	SD	Min.	Max.
Arthritis among older adults	6,053	42.29	42.30	9.78	9.5	77.5
Binge drinking within past 30 days	6,640	9.55	9.10	4.62	0.6	42.3
Cholesterol checked in past 5 years	4,033	93.75	94.60	3.78	67.3	99.9
Colorectal cancer screening	4,023	69.51	70.40	10.01	29.4	94.0
Current smoking	7,264	15.01	13.90	6.75	2.0	65.6
Diabetes screening within past 3 years	4,378	69.66	68.10	9.60	10.2	100.0
Disability status	7,058	38.44	38.00	9.76	6.5	83.0
Duration of caregiving	3,414	72.66	73.00	7.51	0.0	100.0
Eating 2+ fruits daily	3,893	33.90	34.10	7.39	10.5	64.2
Eating 3+ vegetables daily	3,532	15.56	15.00	4.68	2.2	44.2

Ever had pneumococcal vaccine	7,579	56.41	59.40	14.53	12.0	85.8
Expect to provide care	3,552	16.20	15.70	5.44	0.0	100.0
Fair/poor health among adults with arthritis	5,365	34.88	32.50	10.26	8.2	87.2
Fall with injury within last year	2,587	10.60	10.30	3.01	1.7	31.8
Frequent mental distress	7,072	10.97	10.40	3.94	0.9	43.8
Functional difficulties associated with cognitive decline	3,403	39.19	37.80	13.52	0.0	100.0
High blood pressure ever	4,057	53.08	52.80	10.92	15.5	94.7
Influenza vaccine within past year	8,062	51.74	51.70	10.88	8.8	85.4
Intensity of caregiving	3,367	33.39	32.30	10.15	0.0	100.0
Lifetime diagnosis of depression	7,436	17.93	17.30	5.84	1.4	57.3
Mammogram within past 2 years	2,541	73.63	73.50	6.52	42.2	95.8
Need assistance due to cognitive decline	3,373	33.75	32.10	12.49	0.0	100.0
No leisure-time physical activity	7,997	30.89	30.10	7.57	4.7	70.5
Obesity	7,891	34.26	33.90	7.62	3.7	72.4
Oral health/tooth retention	4,124	72.50	73.90	10.94	23.8	99.5
Pap test within past 3 years	2,340	58.03	59.40	17.44	4.6	94.4
Prevalence of sufficient sleep	4,131	66.47	67.20	7.71	30.4	90.6
Provide care for friend/family member	3,745	22.08	22.10	5.65	0.0	71.4
Provide care for someone with cognitive impairment	3,170	21.06	23.00	9.60	0.0	97.6
Self-rated fair/poor health	7,944	25.75	23.50	8.88	5.2	75.5

Self-rated good/excellent health	8,200	74.57	76.60	8.96	33.0	98.8
Severe joint pain among adults with arthritis	3,522	32.27	29.75	11.61	8.5	70.8
Subjective cognitive decline/memory loss	**3,902**	**11.67**	**11.30**	**3.93**	**0.0**	**64.4**
Taking medication for high blood pressure	3,727	87.46	88.40	5.71	63.6	99.6
Talked with healthcare professional about cognitive decline	3,458	44.30	44.70	10.88	0.0	100.0
Up-to-date vaccines/screenings	2,410	35.01	35.00	10.16	6.5	66.0
Up-to-date vaccines/screenings	2,392	30.89	30.10	12.42	4.7	69.0

Table 1. Descriptive Statistics of Health-Related Measures Among Older Adults

This table summarizes the distribution of reported health-related measures among older adults in the CDC dataset. For each health topic, n represents the number of non-missing observations. The mean and median describe the central tendency of the reported values, while the standard deviation (SD) measures the variability between observations. The minimum and maximum values represent the observed range. Most measures are reported as percentages, while measures such as *Recent activity limitations* and *Physically unhealthy days* are reported as the mean number of days.

Interpretation:
The descriptive statistics show substantial variation across the health measures. For example, the average prevalence of arthritis was 42.29%, while current smoking averaged 15.01% and frequent mental distress averaged 10.97%. Some preventive-health measures had relatively high average values, including cholesterol screening (93.75%) and taking medication for high blood pressure (87.46%). In contrast, behaviors such as eating three or more vegetables daily (15.56%) and binge drinking (9.55%) had lower average reported values. The differences between the mean and median for several measures also suggest that

some variables may have skewed distributions or unusually high or low observations. Overall, these statistics provide an initial description of the health measures and help identify variables that may warrant further statistical analysis.

Age group	n	Mean SCD (%)	Median (%)	SD (%)	Minimum (%)	Maximum (%)
50–64 years	1,286	11.14	10.70	3.99	0.0	39.7
65 years or older	1,260	12.20	11.80	4.21	0.0	64.4

The 65-years-or-older group had a higher mean reported SCD prevalence than the 50–64 group (12.20% vs. 11.14%). The median was also higher in the 65+ group (11.80% vs. 10.70%). The 65+ group showed slightly greater variability, with a standard deviation of 4.21 compared with 3.99 for the 50–64 group. The maximum value was substantially higher in the 65+ group, suggesting the presence of more extreme observations.

Number of Features Where Sample Was an Outlier	Number of Samples	Percentage
0	880	95.34%
1	43	4.66%
2	0	0.00%
3	0	0.00%
Total	**923**	**100%**

To determine whether outliers occurred in the same samples across different features, I compared Current Smoking, Lifetime Diagnosis of Depression, and Subjective Cognitive Decline using the same state/year/age-group estimates. Among the 923 matched samples, 880 were not outliers for any feature, while 43 were outliers for exactly one feature. No samples were identified as outliers for two or more features.

This suggests that outliers occur in different samples across these features. Therefore, removing an entire sample based on an outlier in one feature could unnecessarily remove valid observations for other features. For this reason, outliers should be handled separately for each feature during feature-specific analysis.